

CO2-AT1 (Analytical problem solving)

10 Use case Diagram - hospital management system.

d) Actors

- patient
- Doctor
- Administrator

b) Use cases

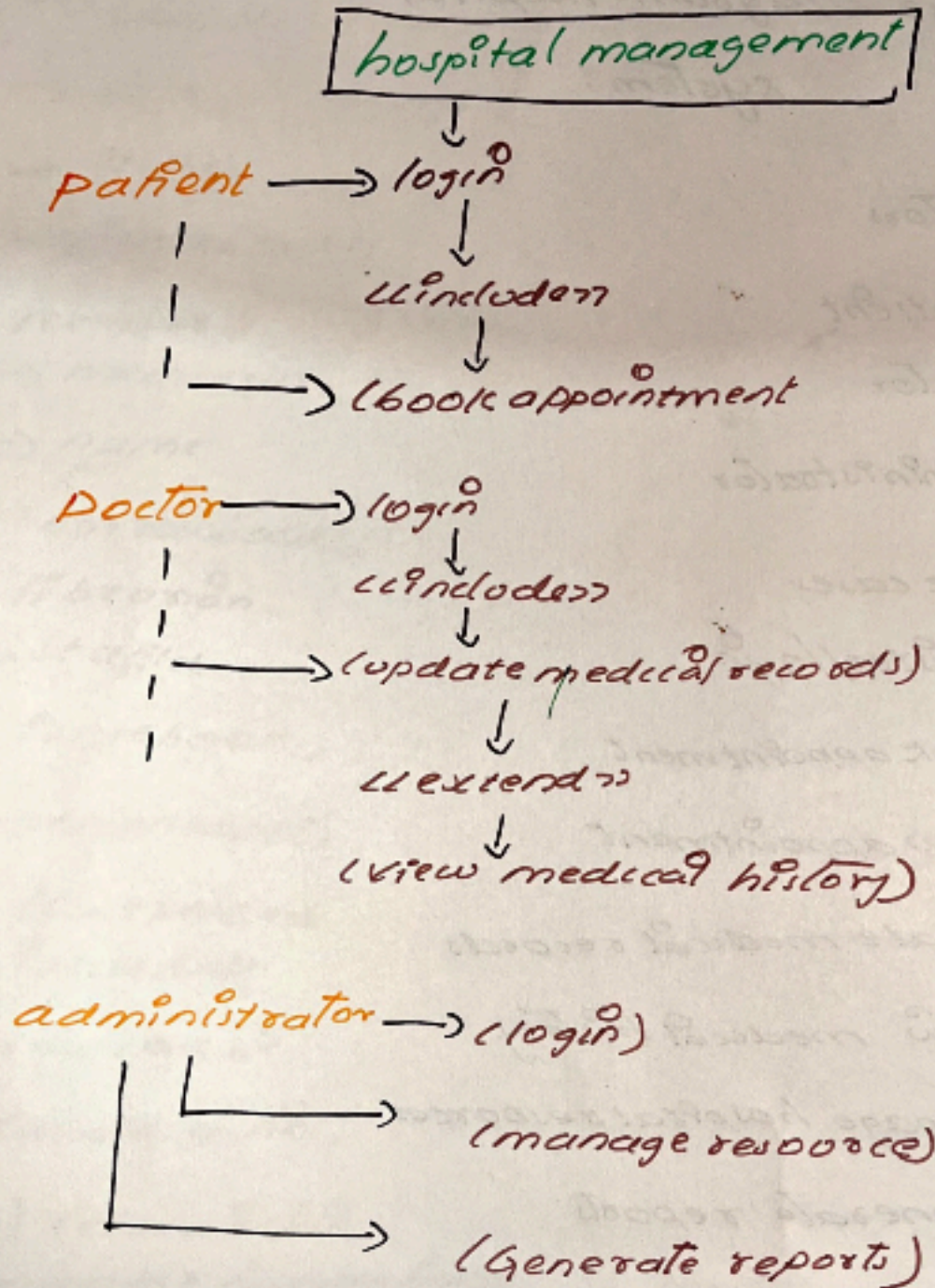
- Register/login
- Book appointment
- View appointment
- Update medical records
- View medical history
- Manage hospital resources
- Generate reports

c) Relationships

- <<include>> login → book appointment
- <<include>> login → update medical records
- <<extend>> view medical history → update medical records

→ Generalization - user - patient doctor, administrator

use case diagram



2. Class diagram - library management system

a) classes

book

→ bookId

→ title

→ author

→ issuebook()

member

→ memberId

→ name

→ borrowbook()

librarian

→ staffId

→ issuebook()

→ returnbook()

Issuerecord

→ issuedate

→ returndate

→ calculatefine()

b) relationship.

→ member borrowbook

→ librarian manages Issuerecord

→ Issuerecord contains book and member.

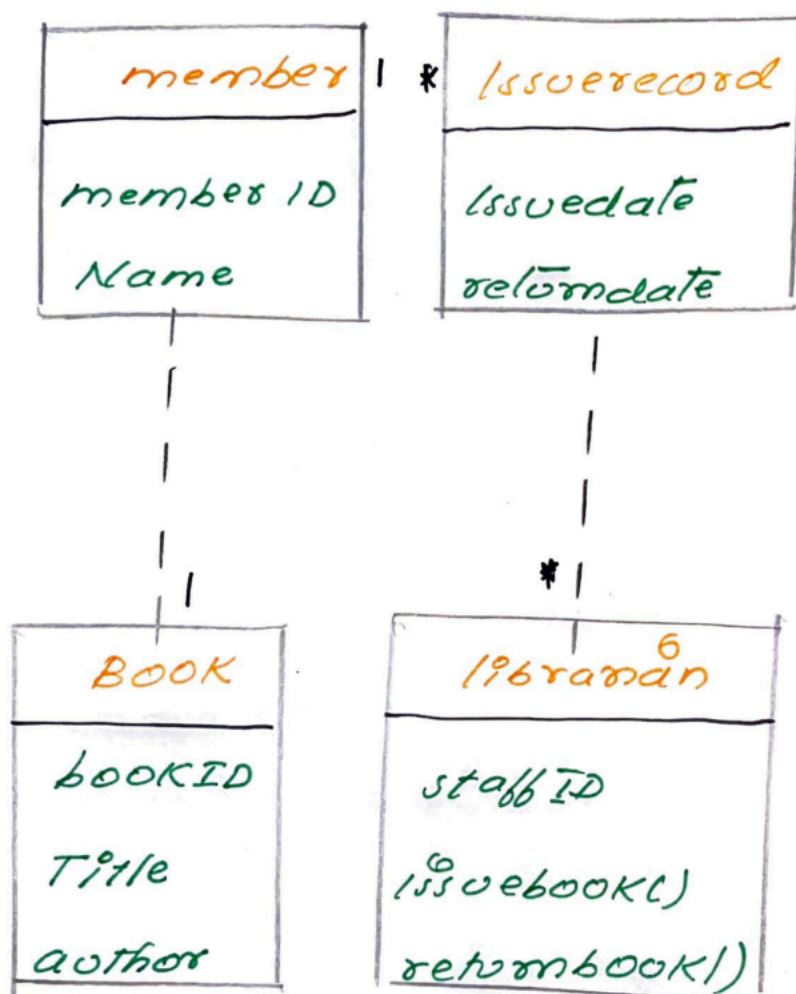
c) multiplicity

→ member 1..* Issuerecord

→ book 1..* Issuerecord

→ librarian 1..* Issuerecord

d) class diagram



3. sequence diagram. online food ordering system.

a) objects

→ customer

→ food app

→ restaurant

→ payment gateway

b) message flow

→ select food

→ place order

→ process payment

→ confirm payment

→ confirm order

c) lifelines

→ customer

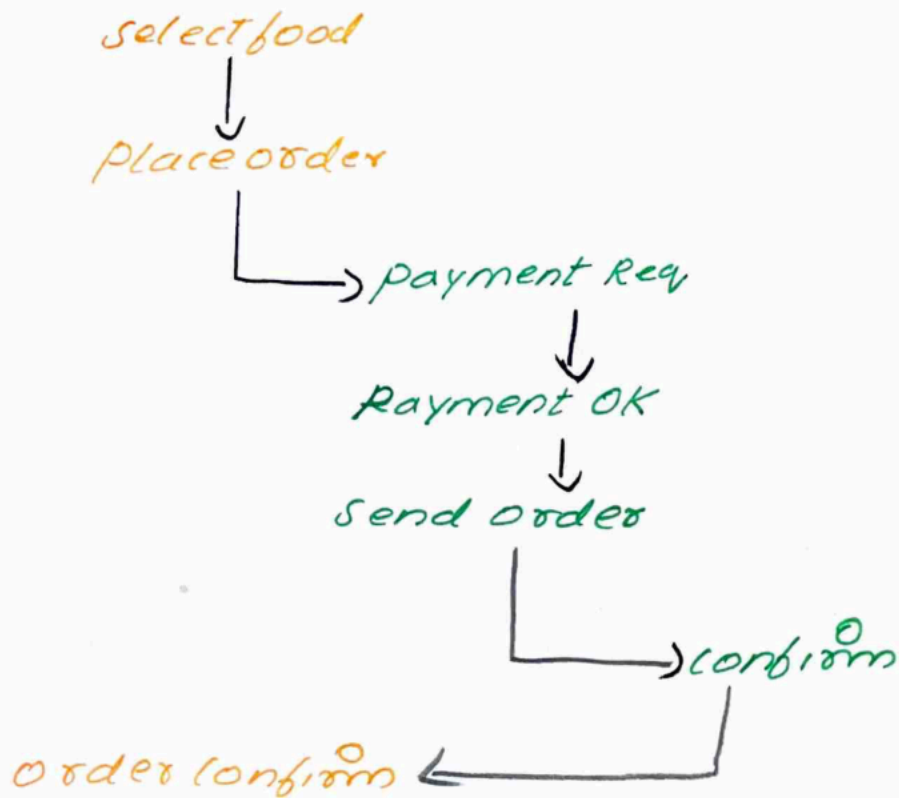
→ food app

→ restaurant

→ payment gateway.

d) sequence diagram.

customer foodapp payment restaurant



4. Activity diagram-student admission system

a) activities

- submit application
- Verify document
- pay fees
- confirm admission

b) Decision

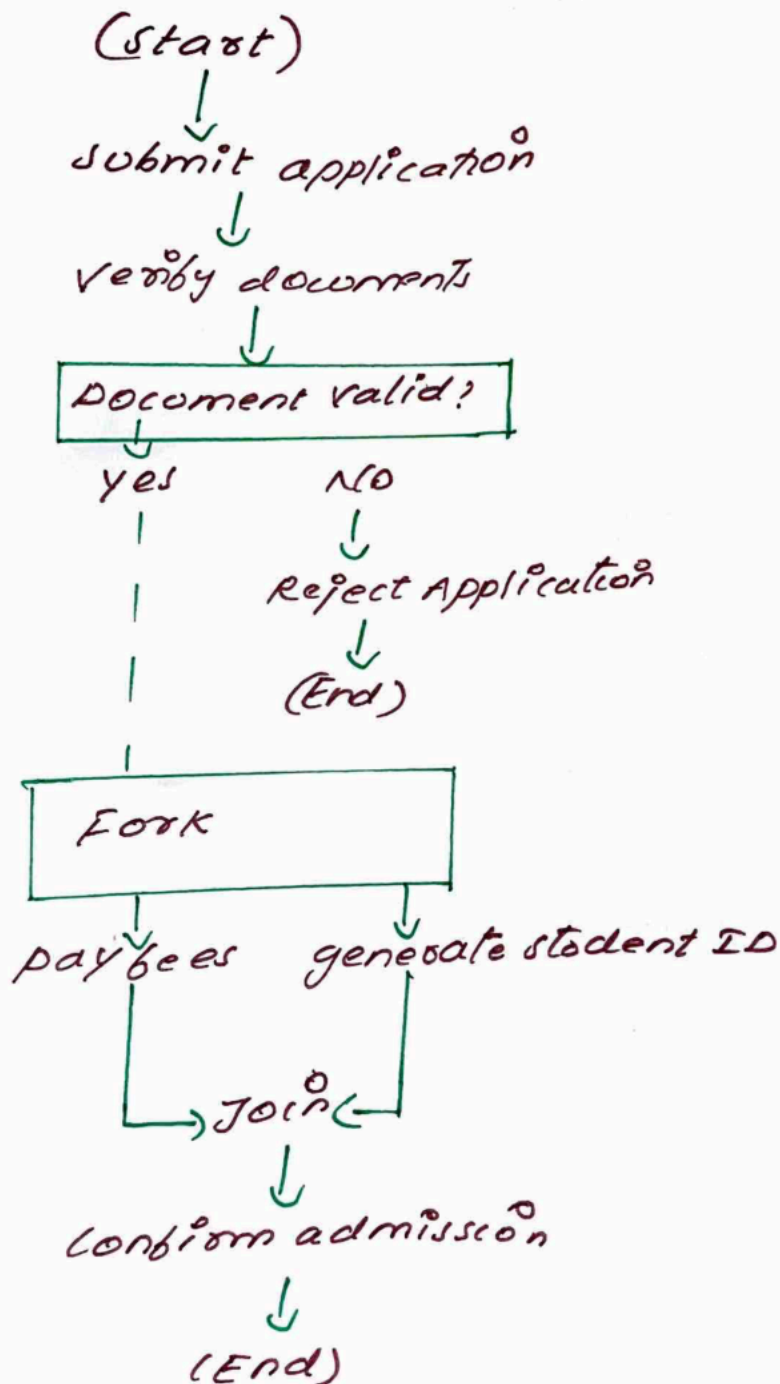
Documents Valid?

c) Parallel process

fee payment

ID Generation

d) activity diagram



5.

state diagram - vending machine

a) states

→ Initial

→ Idle

→ product selected

→ payment received

→ product dispensed

→ Transaction completed

→ final

b) Events

→ select product

→ Insert money

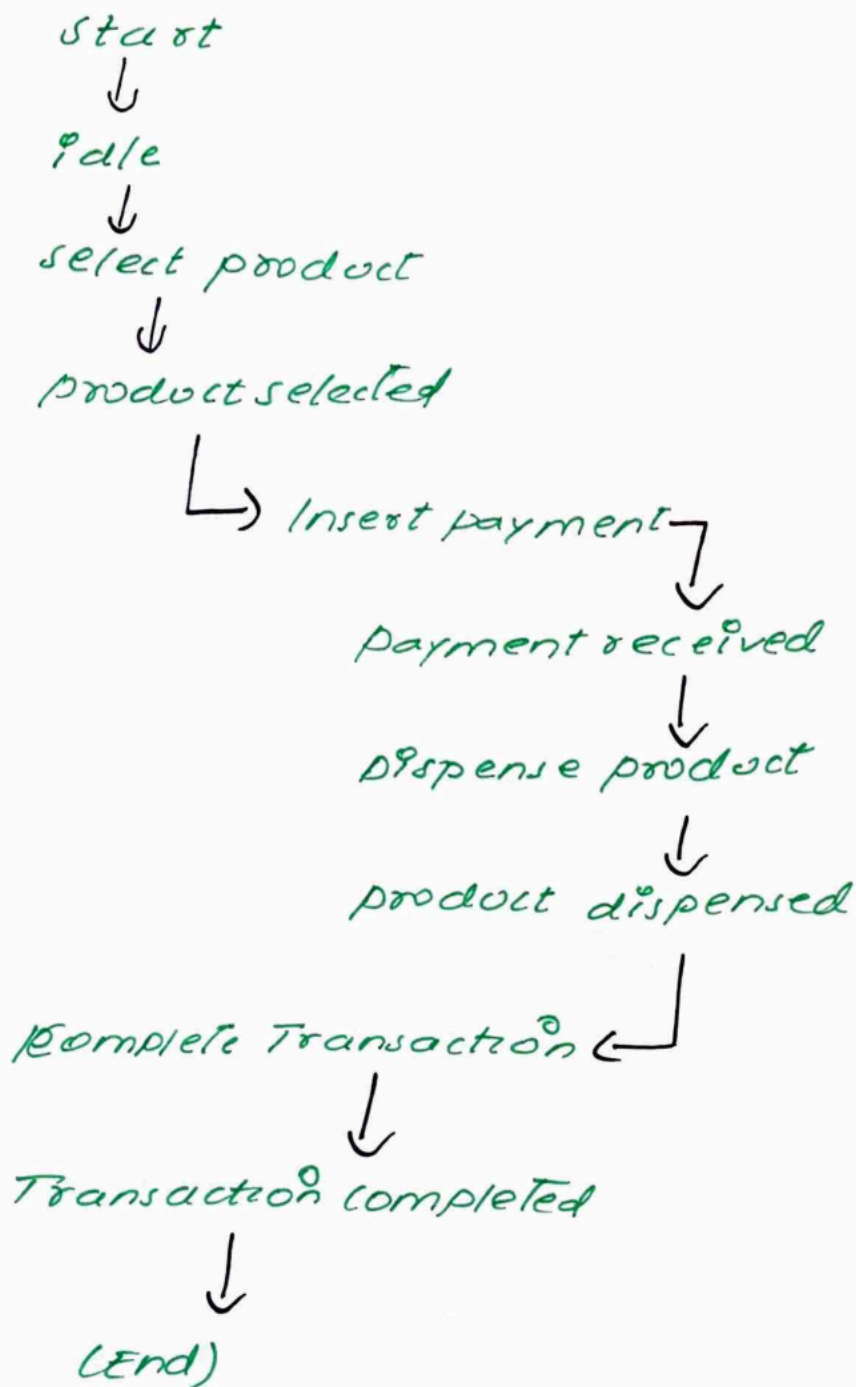
→ Dispense product

→ Finish Transaction

c) Initial & final

→ Included.

d) state diagram.



6. component diagram-online learning platform

a) components

- user management
- course management
- assessment module
- content delivery

→ Notification service

b) Interface

→ login API

→ course API

→ Assessment API

→ notification API

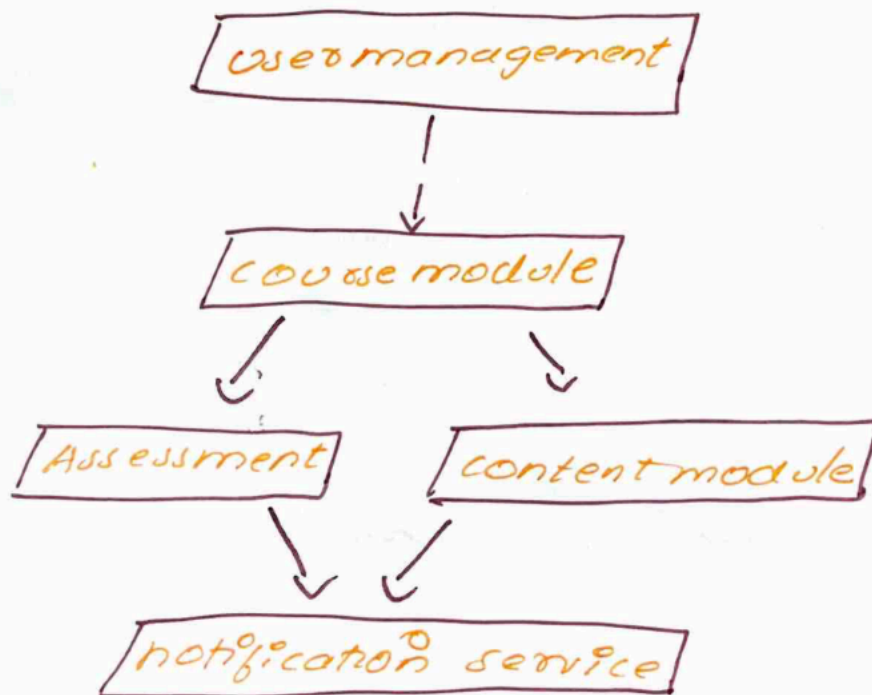
c) Dependencies

→ User → course

→ course → assessment

→ Assessment → notification

d) component diagram



7. Deployment diagram - smart traffic monitoring system.

a) hardware nodes

- Roadside sensors
- IoT gateway
- cloud server
- database server
- web server
- Traffic Authority Computer

b) software components

- sensor software
- data collector
- Traffic monitoring application
- database
- dashboard
- Alert service

c) communication links

- sensors → gateway
- gateway → cloud
- cloud → database
- cloud → dashboard
- dashboard → Traffic authority

d) deployment diagram

